Architecture Overview (PDF)

Architecture Overview ? TechNova Platform

System Architecture

The TechNova Platform follows a microservices architecture deployed on AWS EKS (Kubernetes). The system is divided into the following services:

Core Services

1. API Gateway (Kong) ? Routes external traffic, handles rate limiting and authentication.
2. User Service ? Manages user accounts, profiles, and authentication tokens.
3. Billing Service ? Processes payments via Stripe integration. Handles subscriptions, invoices, and payment retries.
4. Document Service ? Stores and indexes enterprise documents. Integrates with our search engine.
5. Notification Service ? Sends emails (via SendGrid), Slack messages, and in-app notifications.

Data Stores

- PostgreSQL 16 ? Primary database for user data, billing records, and document metadata.
- Redis 7 ? Session cache, rate limiting counters, and pub/sub for real-time features.
- Elasticsearch 8 ? Full-text search for documents and audit logs.
- Amazon S3 ? Object storage for uploaded files and document assets.

Infrastructure

- Kubernetes (EKS) ? Container orchestration for all services.
- Terraform ? Infrastructure as Code for all AWS resources.
- GitHub Actions ? CI/CD pipeline for automated testing and deployment.
- Datadog ? Monitoring, logging, and alerting.

Platform Tech Stack (Official Reference)

When employees ask "What is the tech stack?" or "What technologies does TechNova use?", refer to this section.

Application Layer

Component	Technology	Version	Notes
Customer web app	React	18.x	TypeScript, Vite build
Admin console	React	18.x	Internal operators
Public API	FastAPI	0.11x	Python 3.11 async
Background workers	Celery	5.x	Redis broker
API Gateway	Kong	3.x	Rate limits, JWT validation

Data and Search Layer

Component	Technology	Version	Notes
Primary OLTP database	PostgreSQL	16	RDS Multi-AZ
Cache / sessions	Redis	7	ElastiCache
Document search index	Elasticsearch	8	Clustered, ILM policies
Object storage	Amazon S3	?	SSE-KMS encryption
Message queue	AWS SQS	?	Async jobs

Infrastructure and DevOps

Component	Technology	Notes
Cloud provider	AWS	us-east-1 primary
Orchestration	Kubernetes (EKS)	1.29+

IaC	Terraform	Modules per service
CI/CD	GitHub Actions	OIDC to AWS

Secrets	AWS Secrets Manager	No secrets in git
Observability	Datadog	Metrics, logs, APM
Incident paging	PagerDuty	On-call rotations

Integration Partners

- Stripe ? Billing and subscriptions
- SendGrid ? Transactional email
- LaunchDarkly ? Feature flags
- Okta ? SSO for enterprise customers

Tech Stack Summary (Quick List)

The TechNova tech stack includes: React 18, TypeScript, Vite, Python 3.11, FastAPI, PostgreSQL 16, Redis 7, Elasticsearch 8, Amazon S3, Kong, Kubernetes (EKS), Terraform, GitHub Actions, Datadog, and PagerDuty.

Service Dependency Map

Design Principles

1. Loose coupling: Services communicate via REST APIs and message queues (SQS).
2. 12-factor app: All configuration via environment variables, stateless processes.
3. Graceful degradation: If the notification service is down, core payment processing continues.
4. Defense in depth: Multiple layers of authentication and authorization.

Non-Functional Targets

- API Gateway p95 < 200 ms
- Billing availability 99.95%
- RPO 5 minutes (database PITR), RTO 1 hour (regional failover runbook)

Related Documentation

- api_documentation.md ? REST endpoints
- monitoring_guide.md ? SLOs and dashboards
- database_operations.md ? PostgreSQL operations
- deployment_runbook.md ? release process